

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

SGCC Missingness Strategy & Audit Report

1. Missingness Audit

- **100% Missing Customers:** 5 (FLAG distribution: {1: 3, 0: 2})
- **99-100% Missing:** 8
- **Partial Missingness (>0, <100%):** 42367

2. Missing Gap Distribution

Across the dataset, missing values occur in contiguous blocks (gaps):

- **Total gaps:** 234308
- **Median gap length:** 2.0 days
- **95th Percentile gap length:** 400.0 days
- **Maximum gap length:** 1034 days
- **Gaps > 7 days:** 37656
- **Gaps > 30 days:** 26307
- **Gaps > 90 days:** 22767

3. Imputation Strategy Justification

Given 26307 communication outages exceeding 30 days, unrestricted `ffill()` artificially carries consumption values across months, destroying seasonality and volatility.

Strategy: `ffill(limit=7)`. We will only forward-fill missing values up to 7 days to smooth brief meter disconnections/errors. Gaps longer than 7 days will remain NaN. Statistical features will use `np.nanmean`, `np.nanstd` directly on observed data, ensuring missingness does not distort mathematical realities.

4. Exclusion Criteria

The 5 completely empty customers will be explicitly dropped from the feature set. Because they lack any consumption history, they provide zero signal.

5. Next ML Steps

Remaining NaNs in the engineered feature dataset (caused by customers having mostly NaNs and unable to form trend/volatility metrics) should be handled via a Scikit-Learn `SimpleImputer` (e.g., median) inside a pipeline fitted STRICTLY on the training set to prevent leakage.